# ⚡ JARVIS AI PERSONAL ASSISTANT

Complete Step-by-Step Installation & Configuration Guide for Any Laptop

**VERSION 2.0 (EXECUTIVE)**

**What is Jarvis?** A fully automated, voice-controlled executive assistant that runs locally on your laptop. Features instant voice app launching, YouTube controls, WhatsApp automated replies, email briefing, real-time stock market Buy/Sell analysis, and an Iron Man 3D holographic workspace HUD.

## 1. Prerequisites

- **Operating System:** macOS (MacBook / iMac), Windows 10/11, or Linux.
- **Python:** Python 3.10 or newer (download from python.org).
- **Free Groq API Key:** Free high-speed AI key from console.groq.com/keys.

## 2. 1-Minute Installation (Step-by-Step)

1. **Extract the Package:** Unzip `Jarvis_Setup_Package.zip` into a folder on your Desktop or Home directory.

2. **Run the Automated Installer:**

**On macOS / Linux:** Open your Terminal, navigate to the folder, and run:

```
chmod +x install.sh
./install.sh
```

**On Windows:** Double-click `install.bat` or run in Command Prompt:

```
install.bat
```

3. **Enter Your Free Groq API Key:** When prompted in the terminal, paste your free key from Groq Console. (If you skip, Jarvis still runs in local offline mode).

4. **Launch Jarvis Voice:** To start listening at any time:

```
source .venv/bin/activate && python jarvis_voice.py
```

## 3. Key Features & Voice Commands

| Voice Command | Feature & Execution |
|---|---|
| `"Jarvis, open VS Code"` | **Instant App Launcher:** Launches VS Code, Chrome, Terminal, or any installed app hands-free. |
| `"Play lofi on YouTube"` | **YouTube Automation:** Directly opens YouTube in your browser and starts playing requested music or topics. |

| Voice Command | Feature & Execution |
|---|---|
| `"Stock market report"` | **Live Market Intelligence:** Connects to Yahoo Finance, checks Nifty, Sensex, Reliance, Nvidia & crypto tickers. |
| `"Should I buy Reliance?"` | **Technical Buy/Sell Analysis:** Calculates short/medium-term moving averages & momentum to issue a *BUY*, *SELL*, or *HOLD* signal. |
| `"Send WhatsApp to Pallavi"` | **WhatsApp Automation:** Auto-triggers WhatsApp Web to compose and dispatch intelligent replies. |
| `"Check unread emails"` | **Executive Briefing:** Summarizes inbox into a 30-second spoken update. |
| `"Volume up" / "Mute"` | **System Controls:** Adjusts audio volume or mutes laptop audio instantly. |

## 4. The Gamechanger: Hybrid Failover (Zero Crashes)

Standard AI voice assistants crash or recite *"Rate limit reached, try again later"* whenever cloud tokens run out. Jarvis features an **Autonomous Cooldown Engine**:

- When Groq hits a token limit (HTTP 429), Jarvis parses the exact cooldown seconds without spamming the server.
- It switches seamlessly to local offline Python code to execute commands without interruption.
- You get 100% uptime, zero crashes, and complete privacy.

## 5. 3D Holographic Iron Man HUD

Double-click `jarvis_workplace.html` in any browser (Chrome/Safari) to open the HUD. It includes:

- Rotating wireframe 3D globe and segmented arc reactor rings.
- Real-time responsive audio wave visualizer synced to microphone input.
- Live stock market ticker panel with green/red gain indicators.

### 🇮🇳 Setup Guide (Tenglish Summary for Followers)

**Followers ki easy ga explain cheyadaniki:**

1. Mee computer lo Python 3.10 install chesukondi (python.org).
2. Jarvis folder open chesi, Mac lo aithe terminal lo `./install.sh` run cheyandi, Windows lo aithe `install.bat` double-click cheyandi.
3. Free Groq API key (console.groq.com nunchi) adiginappudu paste cheyandi.
4. `python jarvis_voice.py` run chesi mic mundhu *"Jarvis, open VS Code"* or *"Stock market report"* ani chepthe automatic ga work avthundi!
5. Holographic futuristic screen kosam `jarvis_workplace.html` ని browser lo open cheyandi.